REACTIVE FUNCTIONALITY

Reactive functionality in robotics refers to the ability of a robot to react to changes in its environment in real time without relying on pre-programmed instructions or a fixed sequence of actions. Reactive robots use sensors to gather information about their surroundings and use this information to generate appropriate actions.

A **behavior** is a mapping of sensory inputs to a pattern of motor actions that are used to achieve a task.

- **Reflexive Behavior:** Reflexive behavior is a type of behavior in which a robot responds automatically to a specific stimulus without any internal processing.
  - **Reflexes** - where the response lasts only as long as the stimulus, and the response is proportional to the intensity of the stimulus.
  - **Taxes** - where the response is to move to a particular orientation.
  - **Fixed-action patterns** - where the response continues for a longer duration than the stimulus.
- **Reactive Behavior:** Reactive behavior is a type of behavior in which a robot responds directly to its environment, using feedback mechanisms to adjust its behavior. Reactive robots use sensors to detect changes in their environment, and they respond to these changes with pre-programmed behaviors.
- **Conscious Behavior:** Conscious behavior is a type of behavior in which a robot has an internal representation of its environment and can reason about its actions. Conscious robots use sensors to gather information about their environment, and they use this information to build a model of the world.

Computational Theory

According to this theory, understanding a cognitive system involves identifying the computational problem that it is solving, the algorithm that it is using to solve the problem, and the physical implementation of that algorithm in the system's hardware.

Level 1: Existence proof of what can/should be done.

- **Computational Level**: This level is concerned with what the system does and why it does it. It involves identifying the goal or function of the system and what problem it is trying to solve.

Level 2: Decomposition of "what" into inputs, outputs, and transformations.

- **Algorithmic Level**: This level is concerned with how the system solves the computational problem identified at the first level. It involves developing a step-by-step algorithm or procedure for performing the computations necessary to achieve the system's goal.

Level 3: How to implement the process.

- **Implementational Level**: This level is concerned with the physical implementation of the algorithm in the system's hardware. It involves understanding how the system's neurons, circuits, or other physical components carry out the computations specified by the algorithm.

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Schema Theory

It is a concept in robotics that refers to a way of representing knowledge in a modular and reusable manner. In this theory, knowledge is organized into small units or building blocks called "schemas," which can be combined and modified to create more complex behaviors.

- A **schema** is used in cognitive science and ethology to refer to a particular organized way of perceiving cognitively and responding to a complex situation or set of stimuli.
  - It has two parts: **the knowledge of how to act and/or perceive** (knowledge, data structures, models) and **the computational process** (the algorithm or procedure) it uses to accomplish the activity.
- Schema theory provides a helpful way of translating the insights from behaviors into an object-oriented programming format.
  - Example: Rana Computatrix, the SENSE and ACT components (feed and flee behavior).
- The **motor schema** represents the template for the physical activity, and the **perceptual schema** embodies the stimulus and any fixed-action delays.

Action-Perception Cycle

The **action-perception cycle** refers to the continuous process of perceiving and acting on the environment. This cycle involves the perception of sensory information from the environment, followed by the formulation of a motor plan to perform an action based on that perception, and then the execution of the action.

- The cycle relied on perception to perceive the potential for action afforded by the world. J.J Gibson referred to his work as an "**ecological approach**" because he believed that perception evolved to support actions and that discussing perception independently of an agent's environment and survival behaviors is silly.
  - **Affordances** are perceivable potentialities of the environment for an action.
- **Perception Systems**
  - ***Direct Perception*** **-** This is the "Gibsonian," or ecological, track of the brain and consists of structures low in the brain that evolved earlier on and account for affordances.
  - ***Recognition*** **-** This ties in with problem-solving and other cognitive activities and accounts for the use of internal models to distinguish "your coffee cup" from "my coffee cup."
- **Innate Releasing Mechanisms (IRM) –** It is the most basic biological model of how behaviors convert perception into action.
  - ***Ways to Acquire Behavior***:
    - **Innate** - to be born with a behavior.
    - **Sequence of innate behaviors** - to be born with a sequence of innate behaviors.
    - **Innate with memory** - to be born with behaviors that need some memory for initialization
    - **Learn** - to learn a set of behaviors.
  - ***Categories of Concurrent Behaviors***:
    - **Equilibrium** (the behaviors seem to balance each other out)
    - **Dominance of one** (winner take all):
    - **Cancellation** (the behaviors cancel each other out)

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- ***Functions of Perceptions:***
  - **Stimulus Detection**: The first function of perception in IRMs is to detect relevant stimuli in the environment. This involves using sensors such as cameras, microphones, or touch sensors to gather information about the surroundings.
  - **Stimulus Classification**: The second function of perception in IRMs is to classify the detected stimuli based on their significance for the robot's behavior. For example, a robot might detect several objects in its environment but only classify one of them as a target for grasping.

Locomotion

In theory, ground robots can move in a variety of ways, either in some biological analog of slither, crawling, or walking or through a purely mechanical motion, such as rolling. In practice, ground robots almost always roll using wheels or tracks.

- **Steering** is the process by which a robot adjusts its motion to achieve a desired trajectory or direction of movement. In robot locomotion, steering involves controlling the robot's motion and direction by adjusting the speed and direction of its wheels, legs, or other locomotion mechanisms.

Holonomic locomotion refers to systems that can move in any direction without changing their orientation. This means that holonomic robots can move in any direction without having to turn first, and they can also rotate in place.

- **Synchro-drive**, where the wheels must turn together, approximating turning in place, though there is some slippage of the wheels on the floor. The wheels are linked through either a drive belt or a set of gears.
- **Omnidirectional (Mercanum)** wheels with rollers that allow the robot to move perpendicular to the main wheel. The robot can go sideways by adjusting the relative speeds of the forward and lateral wheels.

Nonholonomic locomotion refers to systems that require changes in orientation to change direction. This means it can only move in certain directions and not rotate in place.

- **Ackerman steering** is how a car is steered; the wheels in front turn to guide the vehicle. This steering refers to the mechanism that adjusts the wheels so that the wheel on the inside of the turn is rotated more than the wheel on the outside of the turn in order to compensate for a slight difference between the inside and outside wheel diameters.
- **Skid steering (Differential steering)** is how a tank or bulldozer is steered. The tracks on each side can be controlled independently. If the robot is to turn left, the left track slows down while the right track speeds up.

Biomimetic Locomotion

The central concept in biomimetic locomotion is that there is a periodic or repetitive motion, either a vibration of the body or an oscillation of limbs.

- **Crawling** occurs when the agent overcomes friction through longitudinal vibration or movement. An example in nature is a caterpillar, which contracts along the length of its body to move forward, creating a slow sinusoidal wave of motion.

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- **Sliding** occurs when the agent overcomes friction through transverse vibrations or movements. An example in nature is a snake that moves in a sinusoidal side-to-side pattern using its scales to increase friction, enabling pivoting. A snake robot uses transverse vibration to move.
- **Running** occurs when the agent overcomes kinetic energy with an oscillatory movement of a multi-link pendulum that leads to a horizontal motion.
- **Jumping** is another way the agent can overcome kinetic energy with the oscillatory movement of the legs. In this case, the oscillations produce a predominately vertical motion.

Legged Locomotion

These robots mimic the movement and behavior of animals that use legs to move, such as dogs, horses, and insects. These robots typically have multiple legs that are articulated and controlled to generate motion and stability.

- **Static Balance -** Whether an agent is balanced as it moves relies on the notion of a support polygon.
  - **Static stability** refers to the ability of a robot to remain upright and stable without the need for active control. This type of stability is achieved by having a wide support base, which reduces the likelihood of the robot tipping over.
- **Dynamic balance** refers to the ability of a legged robot to maintain stability through active control and movement.
  - **Dynamic stability** refers to the ability of a robot to maintain balance and stability by actively adjusting its motion and behavior. This type of stability is achieved using sensors and control systems to monitor the robot's motion and make real-time adjustments.

Gait refers to a limb movement pattern used by a legged robot or an animal to move or walk. A gait is characterized by the sequence, timing, and coordination of the movement of each leg.

- **Walk**: This is a slow and steady gait in which at least one foot is always in contact with the ground.
- **Trot**: This is a two-beat gait in which diagonal legs move together. For example, the front right leg and the rear left leg move together, followed by the front left and rear right leg.
- **Gallop**: This is a four-beat gait in which one foot lands ahead of the others, and then the other three feet land quickly.
- **Bound**: This is a hopping gait in which both legs on one side of the robot or animal move together, followed by both legs on the other side.
- **Crawl**: This is a slow, creeping gait in which the legs move in a coordinated manner, but the robot or animal remains close to the ground.
- **Climb**: This is a gait used for climbing surfaces in which the robot or animal alternately lifts and lowers its legs in a coordinated manner.

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